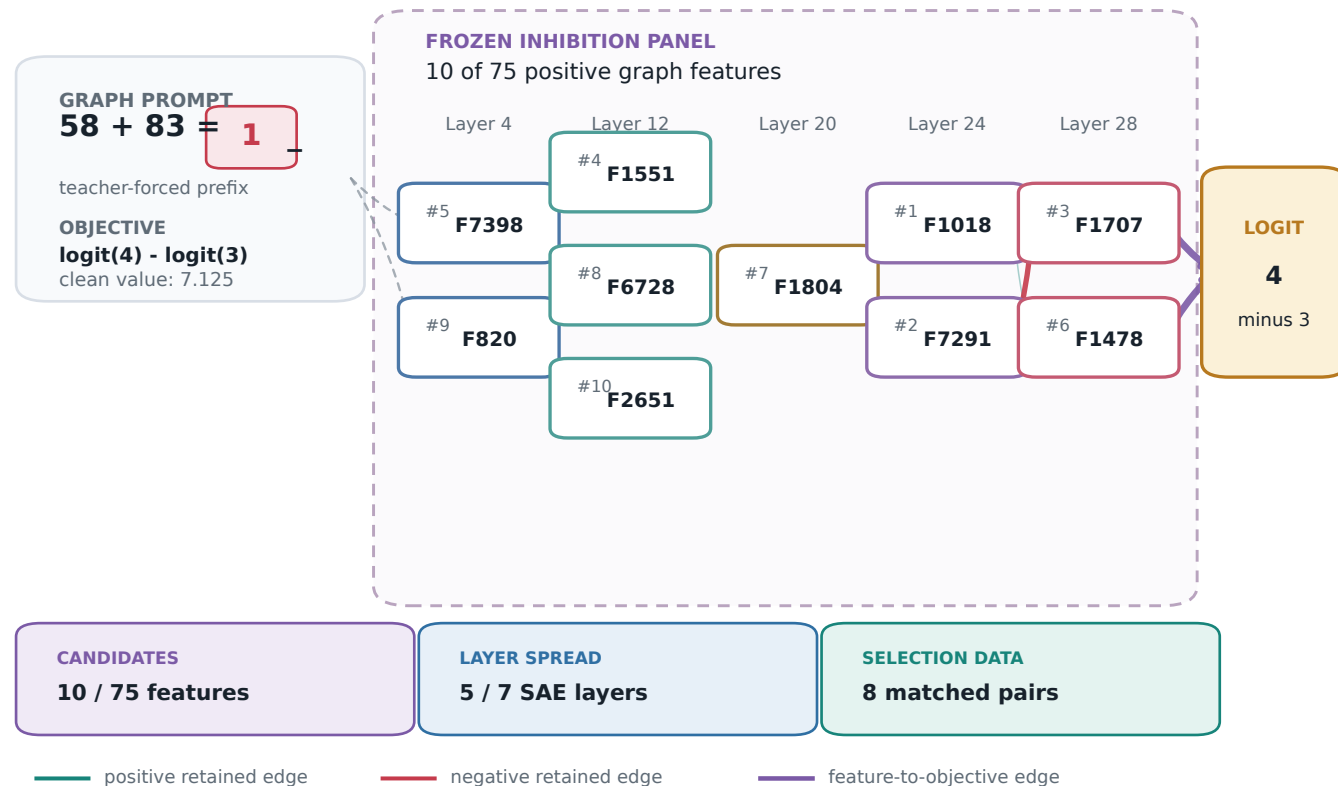


Arithmetic feature inhibition affects digit state, not carry specifically

Attribution-guided TopK SAE panel and frozen matched-control confirmation

A Candidate panel from one carry graph



B Frozen causal specificity test

INTERVENTION

INHIBIT THE SAME 10 LATENTS

$z_j < 0$ at the final token; reconstruction error preserved

CARRY TARGETS

24 held-out cases

NO-CARRY CONTROLS

24 matched controls

MEAN CHANGE IN CORRECT-MINUS-CONTRAST GAP

carry -0.141

control -0.109

FROZEN TOP-10 CONFIRMATION

Paired specificity

-0.031

Bootstrap 95% CI

[-0.089, +0.026]

21/24 carry effects negative

10/24 beat control

Token flips

0 carry | 0 control

PREDECLARED CARRY-SPECIFICITY RULE: NOT MET

Interpretation: inhibition produced a reproducible directional arithmetic effect, but the matched no-carry effect was nearly as large.

The panel is therefore carry-associated at discovery, not an identified abstract carry variable. Solid arrows are retained graph edges; dotted input paths compress displayed token edges.